COURSE GUIDE & SYLLABUS · GRADES 3–5 · AGES 8–10

Scratch (Gr 3–5) Intermediate: Story, Quiz & Music Studio

Pattern art, choose-your-path stories, quiz shows and music machines — coded by you.

The second course in the Scratch pathway. Learners deepen loops (nested loops, variables in loops), build interactive stories with messages and questions, make quiz games with operators, random numbers, scores and timers, and compose music with notes and keyboard instruments. Four projects and a capstone interactive show. Every lesson has a video, a build-along, Code Lab challenges checked automatically, a worksheet and a teacher lesson plan.

What learners will be able to do

- Use nested loops and variables inside loops to generate patterns
- Coordinate characters and scenes with broadcast messages
- Collect input with ask and answer, and branch with if-else
- Use operators to calculate and check answers
- Generate random questions and keep score with variables
- Use the timer to add time limits
- Compose melodies and rhythms with notes, beats and loops
- Turn keys and sprites into instruments and sync animation to music
- Plan, build, test and present an original interactive show

How every lesson works

Watch	A short animated explainer video with captions (and an optional read-aloud).
Learn	Learn cards with the real coding blocks, and a step-by-step build-along.
Try it (Code Lab)	Code in the built-in Scratch-style Code Lab with a live stage. Challenges are checked automatically and give a clear next step; hints unlock one at a time.
Puzzles & quick checks	Robot puzzles and questions that build understanding, with instant feedback.
Probe	Debugging and “new situation” tasks to show real understanding.
Projects	Each module ends with a creative project built in the Code Lab and reviewed by the teacher with a rubric.
Printables	A worksheet for every lesson, a project brief for every project, and a lesson plan with answer keys for the teacher.

Scope and sequence

#	Lesson	Big idea	Learners build
1	**Loops Inside Loops: Pattern Art** Use a loop inside another loop (a nested loop) to draw repeating patterns such as flowers and rows of shapes.	Nested loops	Draw a flower of squares
2	**Variables in Loops: Spirals and Growing Patterns** Change a variable each time round a loop to make patterns that grow, like spirals, and to count.	Variables inside loops	Square spiral
3	**Project: Pattern Art Gallery** Plan, build and share an art gallery of at least two generated patterns using nested loops and variables.	Project: pattern art	Project: Pattern Art Gallery
4	**Scenes and Messages: Telling a Story** Tell a story in scenes: use broadcast messages so characters take turns, and change backdrops between scenes.	Broadcasts and timing	A two-character conversation
5	**Ask and Answer: Stories That Listen** Use “ask and wait” and the answer reporter to personalise a story and let the reader choose what happens next.	Input and branching	A story that knows your name
6	**Project: Choose-Your-Path Story** Plan, build and test an interactive story in scenes, where the reader’s answers change what happens.	Project: interactive story	Project: Choose-Your-Path Story
7	**Operators: Maths in Code** Use operator blocks (+ − × ÷ and join) to calculate, and check a typed answer against the right result.	Operators and comparison	A times-table checker
8	**Random Questions, Score and Timer** Generate random quiz questions, keep score in a variable, and add a time limit with the timer.	Random, variables and timing	A random times-table quiz
9	**Project: Quiz Show** Plan, build and test a quiz game with at least five questions, a score, feedback and a final message.	Project: quiz game	Project: Quiz Show
10	**Notes and Melodies** Play notes and melodies with the music blocks, using note numbers, beats and loops for rhythms.	Sequences in music	Play a tune
11	**Keyboard Instruments** Turn keys and sprites into a playable instrument with events, and sync animations to the music with messages.	Events and parallel scripts	Build a keyboard piano
12	**Capstone: My Interactive Show** Design, build, test and present an interactive show that combines story, quiz or music with events, loops, variables and messages.	Capstone: everything together	Project: My Interactive Show (capstone)

Modules

Module 1 · Pattern art
Nested loops and variables inside loops make flowers, rows and spirals. Project: Pattern Art Gallery.

Big question: How can a few rules create beautiful art?

Module 2 · Interactive stories
Characters take turns with messages, scenes change, and stories listen to the reader. Project: Choose-Your-Path Story.

Big question: How can a story change depending on the reader?

Module 3 · Quiz and maths games
Operators, random questions, scores and timers. Project: Quiz Show.

Big question: How does a computer check if you're right?

Module 4 · Music machines
Notes, melodies, rhythms and keyboard instruments, then a capstone interactive show.

Big question: How can code make music and bring everything together?

Assessment

Coding challenges are checked on the server by running the learner's program. Quick checks give instant feedback. Written answers and the four projects are reviewed by the teacher against 4-level rubrics (Beginning · Developing · Secure · Excellent).

What you need

- Laptop, Chromebook or tablet with a modern browser
- The built-in Kitabash Code Lab (no account needed)
- Headphones for the music lessons
- Optional: Scratch (scratch.mit.edu or the Scratch app)

Before you start

- Scratch (Gr 3–5) Beginner: Scratch Game Makers, or equivalent experience with events, loops, if-then and variables

Safety and privacy

Learners never ask for or share private information in their stories and quizzes (only first names or choices). The Code Lab runs inside Kitabash; sharing to scratch.mit.edu is optional and needs a

parent-approved account for under-13s. Learners under 18 use the AI tutor only with guardian consent.

What comes next

After this course: **Scratch (Gr 3–5) Advanced**, which goes deeper into clones, custom blocks, lists and bigger games.

Kitabash original curriculum. Scratch is a project of the Scratch Foundation in collaboration with the MIT Media Lab; this course is not affiliated with or endorsed by them.